Hum a Beat, Find a Loop: Few-Shot Retrieval of Drum Loops from Beatboxed and Tapped Queries

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1. Problem and Motivation

Music producers and DJs often search large sample libraries for a specific drum groove. Searching by text, such as "funky boom-bap break," is imprecise because tags describe genre and mood rather than where each kick, snare, and hi-hat falls. People usually communicate a groove by performing it instead, for example by beatboxing or tapping on a desk. Our task is to take a short beatboxed or tapped recording and return drum loops that share its rhythm. The **input** is a one- to two-bar query recorded on a laptop microphone. The **output** is a ranked list of the top k matching drum loops from a library. The problem is hard for three reasons. A voice and a drum kit can play the same rhythm with very different sounds. Amateur performers drift in tempo and skip or merge drum parts. And every person beatboxes differently.

2. Related Work and Research Gap

Searching for drum loops by beatboxing was first explored in 2004 [1, 2]. Both systems first convert the query into a sequence of drum hits and then compare that sequence with the library. We call this approach *transcribe-then-match*. Later work showed that the transcription step improves when it is tuned to each user [3]. More recently, TRIA used beatbox and tap recordings to generate new drum audio, using a simple two-band summary of the sound to capture rhythm [4]. General-purpose audio embeddings such as CLAP [5] are widely used for search, but they are trained to connect audio with text descriptions, so they may miss fine rhythmic detail.

Research gap. Deep learning has changed how most audio search works, but searching by beatboxing has not been revisited with modern methods. We do not know whether a model that learns to compare rhythms directly can beat the older transcribe-then-match approach, especially for people the system has never heard. We also do not know whether a few recordings from a new user are enough to adapt such a model to their voice. Our project answers these two questions.

3. Proposed Method

We will train a neural network that turns both queries and drum loops into compact numeric summaries, called embeddings, so that recordings with the same rhythm end up close together and recordings with different rhythms end up far apart (Fig. 1). To search, we compare a query's embedding with every loop in the library and return the closest matches.

Training data. Large sets of matched beatbox recordings and drum loops do not exist, so we will create them. Starting from drum patterns stored as MIDI, we render each pattern two ways: with real drum kits to build the library, and with recorded amateur beatbox sounds to build synthetic queries. We then add realistic imperfections such as timing errors, tempo drift, missing hits, and background noise. Because both sides come from the same pattern, we know exactly which loops should match each query.

Model. We will compare several ways of representing the audio, ranging from simple rhythm features to features from a pretrained music model (MERT [6]). A small encoder trained with a contrastive loss [7] maps these features to embeddings.

User adaptation. For a new user, we record a few examples of their kick, snare, and hi-hat sounds and use them to briefly fine-tune a small adapter [8] on the query side of the model.

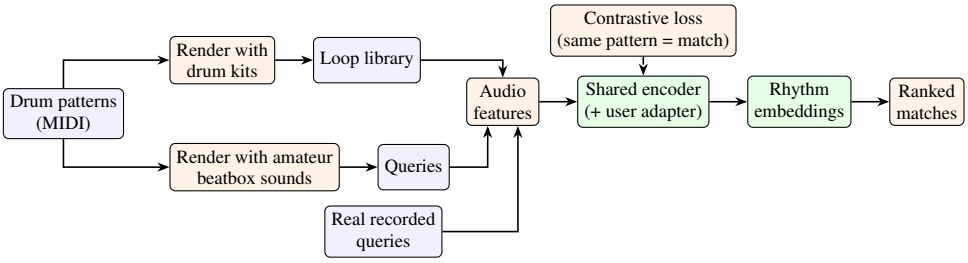


Figure 1: Overview. Library loops and training queries are rendered from the same drum patterns, so we always know which loops should match a query. Real recordings are used for final testing.

4. Datasets and Experiments

Datasets. Drum patterns come from the Groove MIDI Dataset [9] and its expanded version E-GMD [10], which provides the same performances played on many drum kits. We cut the performances into two-bar clips and give each distinct rhythm its own pattern ID. Beatbox sounds come from the Amateur Vocal Percussion (AVP) dataset [11], recorded by 28 people with little

beatboxing experience. Real tap queries come from TapTamDrum [12], and our team will record about 200 real beatbox queries.

Splits. The rhythms, drum kits, and AVP participants used for testing never appear in training, so every test measures performance on unseen patterns, sounds, and voices.

Baselines. We compare against CLAP embeddings, a simple signal-processing method that correlates onset patterns, and a modern transcribe-then-match system that detects and labels each drum hit before matching.

Metrics. Mean reciprocal rank (MRR) is our main metric. We also report recall at 1, 5, and 10 and mean average precision, with bootstrap confidence intervals.

Research questions. RQ1: Does a learned rhythm embedding find the right loop more often than transcribe-then-match and CLAP for unseen users? **RQ2**: Which audio representation works best when moving from synthetic to real queries? **RQ3**: How much do a few enrollment recordings (1, 3, or 5 per drum sound) help a new user?

Ablations. We will vary the query type (beatbox or tap), whether real queries are used for fine-tuning, the amount of tempo drift, missing hits, and noise, and the choice of contrastive loss.

5. Novelty

Our project goes beyond reproducing existing systems in four ways. First, it offers, to our knowledge, the first direct comparison of learned embeddings with the transcribe-then-match approach for this task. Second, it introduces a way to generate realistic training queries from real amateur beatbox sounds and checks them against real recordings. Third, it studies few-shot adaptation to a user's voice for search rather than transcription. Fourth, we will release our data splits as a benchmark.

6. Feasibility and Fallback Plan

All datasets are public. The models are small, pretrained features are computed once and stored, and training fits on a single GPU. The main risk is that synthetic queries may not sound like real ones. Our real-query test set measures this directly, and fine-tuning on real queries can reduce the gap. Estimating tempo from beatboxing is also unreliable, so real queries are recorded with a click track and tempo drift is tested separately. If TapTamDrum patterns cannot be matched to our library, we will test taps on synthetic queries only. If the learned models underperform, the transcribe-then-match system becomes our main system, and the report will explain where and why each approach fails. We can also narrow the scope to one-bar patterns with only kick, snare, and hi-hat.

7. Timeline and Responsibilities

Weeks	Milestones
1–3	Build the loop library, query generator, evaluation code, and first baselines. **Dataset frozen at the end of week 3.**
4–5	Record real queries, build the transcribe-then-match baseline, and train the main models.
6–7	User adaptation, ablations, robustness tests, and error analysis.
8	Final report and presentation.

[Joshua Yu]	Music lead: makes the project's musical decisions (how drum parts are grouped, what counts as the same rhythm, how people beatbox and tap), writes them as clear rules, chooses drum kits, leads recording sessions, checks the audio by ear, and analyzes mistakes.
[Junior Chaj Mejia]	Data pipeline: turns the music lead's rules into code that builds the loop library and the synthetic beatbox and tap queries, organizes all recordings, and runs the user adaptation experiment.
[Anika Nayak]	Builds and trains the AI model that summarizes rhythm, and tries different ways of feeding in the audio.
[Dylan Palmer]	Evaluation: builds the answer key, scoring code, and simpler comparison systems, tests messier recordings, and makes the results tables.

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